

Safety Beyond the Training Data: Robust Out-of-Distribution MPC via Conformalized System Level Synthesis

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Abstract

We present a novel framework for robust out-of-distribution planning and control using conformal prediction (CP) and system level synthesis (SLS). Our method addresses the challenge of ensuring safety and robustness when using learned dynamics models beyond the training data distribution. We first derive high-confidence bounds on model errors using weighted conformal prediction with a learned, state-control-dependent covariance model. These bounds are then integrated into an SLS-based robust nonlinear model predictive control (RMPC) formulation, which performs constraint tightening over the prediction horizon via volume-optimized forward reachable sets. We provide theoretical guarantees on coverage and robustness under distributional drift, and analyze the impact of data density and trajectory tube size on prediction coverage. Empirically, we demonstrate our approach on nonlinear systems of increasing complexity, including a 4D car and a 12D quadcopter, showing improved safety and reliability compared to fixed-bound and non-robust baselines, especially outside of the collected data distribution.

Keywords: safe learning-based control, conformal prediction, MPC, system level synthesis

1. Introduction

Provably-safe motion planning under unknown dynamics is vital for real-world robotic deployment. While traditional planners work well with precisely-known dynamics, real robotic systems often face uncertainty or modeling errors. Data-driven methods such as model-based reinforcement learning (RL) can learn dynamics from data but may exploit model inaccuracies, leading to unsafe or infeasible trajectories. These issues motivate uncertainty-constrained planners that quantify tracking error between true and learned dynamics to ensure robust, safe execution. Recent methods enable safe planning with learned dynamics (Knuth et al., 2021, 2023; Wasiela et al., 2024) by estimating model error bounds and constraining closed-loop behavior within a “trusted domain.” They compute reachable tubes that account for spatially varying model error and reference tracking. However, the trusted domain is typically a bounded neighborhood around the training data, limiting planning far from the data and when start and goal states lie in disconnected regions. In practice, robotics often requires temporarily operating out of distribution (OOD), tolerating degraded performance while maintaining safety. Existing methods are also inefficient, relying on sampling-based planners that do not actively reduce model uncertainty. A key gap is a fast approach that actively shrinks uncertainty to enable less conservative, uncertainty-informed planning.

To address this gap, we propose CP-SLS-MPC, a fast optimization-based feedback planner for robust planning with learned dynamics. CP-SLS-MPC formulates an MPC problem that allows temporary OOD operation while maintaining probabilistic safety. During such operation, online data updates the model error bounds to encourage more cautious behavior, such as keeping greater distance from obstacles. CP-SLS-MPC uses the system level synthesis (SLS) framework Anderson et al. (2019) to jointly design reference trajectories, feedback controllers, and closed-loop reachable tubes. Weighted conformal prediction (WCP) Barber et al. (2023) provides calibrated model error bounds covering true errors with high probability and bounded coverage gap, even OOD, provided the divergence between training and test distributions is limited. These bounds inform SLS tubes, ensuring high-probability safety when tracking learned dynamics. CP-SLS-MPC can also include active uncertainty reduction to guide planning toward lower-error regions. Our contributions are:

- A fast, optimization-based algorithm for robust MPC with learned dynamics which provides finite-sample, probabilistic safety guarantees on the true system via SLS and CP, with state-control-dependent CP-based model error bounds efficiently integrated into the MPC.
- Theoretical analysis of the method, proving that the system remains safe with high probability and bounding the coverage gap in out-of-distribution settings.
- Validation on a simulated nonlinear 4D car and 12D quadcopter, improving safety and prediction accuracy even in OOD scenarios over uncertainty-unaware MPC baselines.

2. Related Work

Predictive Control SLS (Anderson et al., 2019; Chen et al., 2024), equivalent to disturbance-feedback MPC (Goulart et al., 2006), provides a convex parameterization of closed-loop responses for linear time-varying (LTV) systems, enabling joint optimization of nominal trajectories and feedback. Online feedback optimization reduces conservativeness and improves performance under bounded or stochastic disturbances Bartos et al. (2025). Extensions to nonlinear dynamics decompose control into a nominal trajectory and an LTV error system, allowing joint optimization of affine feedback and linearization-error bounds Leeman et al. (2025b). SLS is also effective in data-driven settings, where learned closed-loop responses enable robust control with limited models (Xue and Matni, 2021; Furieri et al., 2022). Broadly, robust MPC ensures constraint satisfaction under disturbance Rawlings et al. (2017). Tube MPC Mayne et al. (2005) uses invariant sets around nominal trajectories. Learning-based MPC methods combine uncertainty estimation with planning Hewing et al. (2020), but typically assume simple error models and lack state-dependent bounds.

Planning with Learned Dynamics Most model-based RL methods perform long-horizon planning with learned models but lack safety guarantees Moerland et al. (2023). Some methods estimate reachable tubes for safe exploration Berkenkamp et al. (2016), but these either assume a fixed controller or restrict planning near training trajectories. The most closely related methods Chou et al. (2021) safely plan with learned dynamics using invariant tubes constrained to lie in a “trusted domain” near training data, demanding dense coverage. In contrast, we allow the robot to move beyond the training data while retaining probabilistic safety guarantees. Later work Suh et al. (2023) improves speed via diffusion-guided optimization but lacks formal guarantees. Also closely related is Marques and Berenson (2024), which leverages CP for model error bounding, but unlike our method, cannot extrapolate OOD as it uses exchangeable CP. Other methods aim to improve dynamics model performance in OOD scenarios Lin et al. (2024) but do not focus on planning.

Conformal Prediction (CP) CP (Vovk et al., 2005) is a distribution-free method for constructing prediction regions from data from black-box models, making it well-suited for safe trajectory prediction and planning (Lindemann et al., 2023; Dixit et al., 2023; Sun et al., 2023). Prior work has applied CP to guide motion planning (Muthali et al., 2023). The most closely related methods are Contreras et al. (2025), which triggers a safe fallback policy in OOD scenarios for MPC, and Chee et al. (2024, 2023), which use WCP to address time-series correlations in MPC for tracking fixed trajectories. However, these methods do not use state-dependent uncertainty. In contrast, our method designs trajectories online in OOD settings and explicitly encourages uncertainty reduction, steering the MPC toward regions of lower model error.

3. Preliminaries and Problem Statement

Let $f : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ be the true unknown deterministic nonlinear discrete-time dynamics,

$$x_{k+1} = f(x_k, u_k) = \hat{f}(x_k, u_k) + \varepsilon(x_k, u_k), \quad (1)$$

where $\mathcal{X} \subseteq \mathbb{R}^{n_x}$ is the state space and $\mathcal{U} \subseteq \mathbb{R}^{n_u}$ is the control space. Let $\hat{f} : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ denote an approximate learned dynamics model and $\varepsilon : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$ as the dynamics error. We model $\hat{f}$ with a neural network (NN), though our method is agnostic to model class. The goal is to reach a target set $\mathcal{X}_f \subseteq \mathcal{X}$ while satisfying pointwise-in-time state and control constraints $(x_k, u_k) \in \mathcal{F}$, where

$$\mathcal{F} := \{(x, u) \in \mathbb{R}^{n_x+n_u} \mid g_i(x, u) + b_i \leq 0, \quad \forall i \in \{1, \dots, n_c\}\}. \quad (2)$$

We denote $\mathcal{B}^a$ as the unit 2-norm ball in $\mathbb{R}^a$. We denote the Minkowski sum as $A \oplus B := \{a+b \mid a \in A, b \in B\}$. Our method relies heavily on CP and SLS, which we overview next.

Conformal Prediction CP is an uncertainty quantification (UQ) framework that produces prediction sets covering the true outcome with probability $1 - \alpha$, where $\alpha \in (0, 1)$ is a user-selected miscoverage rate (Vovk et al., 2005). Traditional CP assumes statistical exchangeability, meaning the data is identically distributed and all permutations of the data ordering are equally likely. This limits its use when spatio-temporal distribution drifts occur, e.g., when a robot plans far from its training data. To address this, we use non-exchangeable (weighted) CP (WCP), which relaxes the exchangeability assumption while retaining similar coverage guarantees (Barber et al., 2023).

Given a calibration dataset, e.g., for our work $\mathcal{D}_{\text{calib}} := \{(x_i, u_i, f(x_i, u_i))\}_{i=1}^n$ with n points, CP and WCP use a non-conformity score function $s : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ to compute an *uncalibrated* notion of uncertainty, i.e., lacking probabilistic coverage guarantees, where a larger non-conformity score indicates greater disagreement between the input (x_i, u_i) and the target $f(x_i, u_i)$. Then defining $s_i := s(x_i, u_i, f(x_i, u_i))$, we compute the $(1 - \alpha)$ -quantile of the empirical distribution

$$S := \sum_{i=1}^n \tilde{w}_i \delta_{s_i} + \tilde{w}_{\text{test}} \delta_{+\infty}, \quad (3)$$

where δ_{s_i} refers to the Dirac delta function centered at s_i and $\tilde{w}_i$ are weights normalized such that $\tilde{w}_{\text{test}} + \sum_{i=1}^n \tilde{w}_i = 1$. All weights $\{\tilde{w}_i\}_{i=1}^n$ and $\tilde{w}_{\text{test}}$ lie within the interval $[0, 1]$. We denote the $(1 - \alpha)$ -quantile of S as $s_{1-\alpha} := Q_{1-\alpha}(S)$ and define the prediction set at test point $(x_{\text{test}}, u_{\text{test}})$ as

$$\mathcal{C}(x_{\text{test}}, u_{\text{test}}; s_{1-\alpha}) := \{x \in \mathcal{X} \mid s(x_{\text{test}}, u_{\text{test}}, x) \leq s_{1-\alpha}\}. \quad (4)$$

We also define S^i as the distribution of the n -tuple $(s_1, \dots, s_{i-1}, s_{\text{test}}, s_{i+1}, \dots, s_n)$, when we replace the i^{th} calibration point with the test point. Then, the WCP method ensures that

$$\Pr[f(x_{\text{test}}, u_{\text{test}}) \in \mathcal{C}(x_{\text{test}}, u_{\text{test}}; s_{1-\alpha})] \geq 1 - \alpha - \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i d_{TV}(S^i, S), \quad (5)$$

where $d_{TV}(S^i, S)$ is the total variation distance between distribution S^i and S . The weights $\tilde{w}_i$ can be tuned to jointly minimize the coverage gap $\sum_{i=1}^n \tilde{w}_i d_{TV}(S^i, S)$ and the size of the prediction set $\mathcal{C}(x_{\text{test}}, u_{\text{test}}; s_{1-\alpha})$. Intuitively, by weighting calibration scores more similarly distributed to the test non-conformity score, we can reduce the coverage gap. Under the standard exchangeability assumption made in traditional CP, $d_{TV}(S^i, S) = 0$ and the gap is eliminated. WCP is crucial for planning with learned dynamics, as it accounts for (1) distribution shifts between training data and correlated states/controls during execution and (2) visits to OOD regions of the state/control space.

SLS for Trajectory Tracking System Level Synthesis (SLS) (Anderson et al., 2019) is a synthesis tool that can be used to optimize a nominal trajectory and a tracking controller jointly, while also propagating system disturbances over time to compute a overapproximate closed-loop reachable tube. In particular, SLS considers control synthesis for uncertain linear time-varying (LTV) systems

$$x_{k+1} = A_k x_k + B_k u_k + E_k \xi_k \quad (6)$$

where $A_k \in \mathbb{R}^{n_x \times n_x}$, $B_k \in \mathbb{R}^{n_x \times n_u}$, and $E_k \in \mathbb{R}^{n_x \times n_x}$, and $\xi_k \in \mathbb{B}^{n_x}$ is a disturbance. Define a nominal state trajectory $\mathbf{z} = \{z_1, \dots, z_T\} \in \mathbb{R}^{n_x T}$ and control trajectory $\mathbf{v} = \{v_1, \dots, v_T\} \in \mathbb{R}^{n_u T}$ satisfying the nominal LTV dynamics in (7). To track the nominal $(\mathbf{z}, \mathbf{v})$, SLS defines a

disturbance feedback controller (8) and closed-loop state response (9) where $\Phi_{k,j}^u \in \mathbb{R}^{n_u \times n_x}$ and $\Phi_{k,j}^x \in \mathbb{R}^{n_x \times n_x}$ define the closed-loop response of (6) as an affine function of the disturbances $\{\xi_k\}$.

$$z_{k+1} = A_k z_k + B_k v_k \quad (7) \quad u_k = v_k + \sum_{j=1}^{k-1} \Phi_{k,j}^u \xi_j \quad (8) \quad x_k = z_k + \sum_{j=1}^{k-1} \Phi_{k,j}^x \xi_j \quad (9)$$

To ensure that (8) and (9) hold, $\Phi_{k,j}^x$ must satisfy

$$\Phi_{k+1,j}^x = A_k \Phi_{k,j}^x + B_k \Phi_{k,j}^u, \quad \Phi_{k+1,k}^x = E_k. \quad (10)$$

With SLS, we can guarantee that the closed-loop trajectories of (9) and (8) remain in the reachable tubes $\mathcal{R} := \{\mathcal{R}_k \doteq (\mathcal{R}_k^x, \mathcal{R}_k^u)\}_{k=1}^T$, i.e., for all $k \in \{1, \dots, T\}$, $x_k \in \mathcal{R}_k^x$ and $u_k \in \mathcal{R}_k^u$, where

$$\mathcal{R}_k^x := (\bigoplus_{j=0}^{k-1} \Phi_{k,j}^x \mathcal{B}^{n_x}) \oplus \{z_k\}, \quad \mathcal{R}_k^u := (\bigoplus_{j=0}^{k-1} \Phi_{k,j}^u \mathcal{B}^{n_u}) \oplus \{v_k\}. \quad (11)$$

Minimizing a surrogate volume of the reachable tubes can be achieved through the following cost,

$$\tilde{H}_0(\Phi^x, \Phi^u) := \sum_{j=0}^{T-1} \left(\|\bar{P}^{\frac{1}{2}} \Phi_{T,j}^x\|_{\mathcal{F}}^2 + \sum_{k=j}^{T-1} \left(\|\bar{Q}^{\frac{1}{2}} \Phi_{k,j}^x\|_{\mathcal{F}}^2 + \|\bar{R}^{\frac{1}{2}} \Phi_{k,j}^u\|_{\mathcal{F}}^2 \right) \right), \quad (12)$$

where Φ^x and Φ^u are lower triangular block matrices, where the k th block row and j th block column are $\Phi_{k,j}^x$ and $\Phi_{k,j}^u$, respectively, and $\bar{P}, \bar{Q}^{\frac{1}{2}} \in \mathbb{R}^{n_x \times n_x}$, $\bar{R}^{\frac{1}{2}} \in \mathbb{R}^{n_u \times n_u}$ are symmetric positive definite matrices. SLS has been recently extended to control nonlinear dynamics (Leeman et al., 2025b) and robust satisfaction of nonlinear constraints. For nonlinear dynamics $\hat{f}$ of the form (1), we can obtain an LTV approximation the LTV dynamics as $A_k := J_x \hat{f}(z_k, v_k)$, and $B_k := J_u \hat{f}(z_k, v_k)$, are the Jacobians of $\hat{f}$ used to linearize the model around a nominal trajectory $(\mathbf{z}, \mathbf{v})$. Finally, to robustly satisfy nonlinear constraints of the form (2), we can enforce the tightened constraints (13), obtained by linearizing g_i , for all $i \in \{1, \dots, n_c\}$, around the nominal trajectory (7):

$$\sum_{j=0}^{k-1} \left\| \left[\nabla_x g_i(z_k, v_k) \Phi_{k,j}^x, \nabla_u g_i(z_k, v_k) \Phi_{k,j}^u \right] \right\|_2 + g_i(z_k, v_k) + b_i + g_i^{\text{lin}} \leq 0, \quad (13)$$

where g_i^{lin} is a linearization error bound term (details are omitted for clarity; see (Zhan et al., 2025, Eq. (22)) for details). This ensures that the closed-loop dynamics satisfy (2) for all $k \in \{1, \dots, T\}$.

Problem Statement We assume that we are given an offline dataset $\mathcal{D} := \{(x_i, u_i, f(x_i, u_i))\}_{i=1}^N$, where each (x_i, u_i) is drawn from a joint distribution Y . We partition $\mathcal{D}$ into a training dataset $\mathcal{D}_{\text{train}} := \{(x_i, u_i, f(x_i, u_i))\}_{i=1}^{N_{\text{train}}}$ and calibration dataset $\mathcal{D}_{\text{calib}} := \{(x_i, u_i, f(x_i, u_i))\}_{i=1}^{N_{\text{calib}}}$. We will leverage $\mathcal{D}_{\text{train}}$ to train our models – the learned dynamics model $\hat{f}$ (1) and a heuristic model error predictor Σ , described in Sec. 4. Our goal is to 1) estimate calibrated error bounds on the learned dynamics $\hat{f}$ that hold with high probability and to 2) use these bounds within a robust MPC framework to guarantee robust constraint satisfaction with high probability. Specifically, we solve: *Problem 1: Learning and Calibration.* Using $\mathcal{D}_{\text{train}}$, learn an approximate dynamics model $\hat{f} : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{X}$. Using $\hat{f}$ and $\mathcal{D}_{\text{calib}}$, provide a calibrated high-confidence bound $E : \mathcal{X} \times \mathcal{U} \rightarrow 2^{\mathcal{X}}$ on the residual dynamics $\varepsilon : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^{n_x}$, such that $\Pr[\bigcap_{k=1}^T (\varepsilon(x_k, u_k) \in E(z_k, v_k))] \geq 1 - \tilde{\alpha}$, for nominal state (z_k, v_k) and for all reachable state/controls $(x_k, u_k) \in \mathcal{R}_k := (\mathcal{R}_k^x, \mathcal{R}_k^u)$, as defined in (11). Here, $\alpha \in (0, 1)$ is user-specified and $\tilde{\alpha} \in [\alpha, 1)$ absorbs the coverage gap in (5).

Problem 2: Planning. Using the learned model $\hat{f}$ and error bound E of Prob. 1, solve a robust MPC problem for the learned dynamics $\hat{f}$ which ensures that the constraints (2) are satisfied for the true closed-loop dynamics (i.e., under f in (1)) with probability at least $1 - \tilde{\alpha}$ for a single MPC solve and with probability at least $1 - \hat{\alpha}$ for R MPC solves (i.e., $R - 1$ replanning steps), where $\hat{\alpha} \in (0, 1)$.

4. Method

We outline our method, CP-SLS-MPC, which uses WCP for high-probability state-control-dependent model error bounding (Sec. 4.1) and integrates it into an uncertainty-constrained SLS-based robust MPC formulation (Sec. 4.2). Finally, we discuss practical implementation in Sec. 4.3.

4.1. Model Training and Model Error Bounding via Conformal Prediction

To tackle Prob. 1, we first train the dynamics $\hat{f}$ (1) on the training dataset $\mathcal{D}_{\text{train}}$ by minimizing a mean-squared-error loss $\sum_{k=1}^{N_{\text{train}}} \|f(x_k, u_k) - \hat{f}(x_k, u_k)\|_2^2$. Then, for any nominal state $z \in \mathcal{X}$ and control input $v \in \mathcal{U}$, we seek to bound the error of the learned dynamics model $\|\hat{f}(z, v) - f(z, v)\|_2$. To encode state-dependent error, we train a heuristic uncertainty model $\Sigma : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}^{n_x \times n_x}$, a NN, with a multivariate Gaussian negative-log-likelihood (MGNLL) loss (Dasgupta and Hsu, 2007):

$$\mathcal{L}(\Sigma; f, \hat{f}, x, u) = \frac{1}{2} \left((f(x, u) - \hat{f}(x, u))^\top \Sigma(x, u)^{-1} (f(x, u) - \hat{f}(x, u)) + \ln(\det(\Sigma(x, u))) \right). \quad (14)$$

Here, Σ outputs a covariance matrix estimating the model error dispersion at a given state and control. We use this non-conformity score for multi-dimensional regression (Messoudi et al., 2022)

$$s(x_i, u_i, f(x_i, u_i)) = \sqrt{(f(x_i, u_i) - \hat{f}(x_i, u_i))^\top \Sigma(z_k, v_k)^{-1} (f(x_i, u_i) - \hat{f}(x_i, u_i))}. \quad (15)$$

Using (15), we construct the $(1 - \alpha)$ -confidence CP set for (z_k, v_k) , $\mathcal{C}(z_k, v_k; s_{1-\alpha})$ as

$$\mathcal{C}(z_k, v_k; s_{1-\alpha}) := \{x \in \mathcal{X} \mid \sqrt{(x - \hat{f}(z_k, v_k))^\top \Sigma(z_k, v_k)^{-1} (x - \hat{f}(z_k, v_k))} \leq s_{1-\alpha}\}. \quad (16)$$

Note $\mathcal{C}(z_k, v_k; s_{1-\alpha})$ is an ellipsoid centered at the predicted next state $\hat{f}(z_k, v_k)$. Then, (5) ensures

$$\Pr[f(z_k, v_k) \in \mathcal{C}(z_k, v_k; s_{1-\alpha})] \geq 1 - \alpha - \sum_{i=1}^n \tilde{w}_i^k d_{TV}(S^i, S_k). \quad (17)$$

Here, S_k is the empirical distribution as defined in (3), except where we select a set of normalized weights $\tilde{W}^k := \{\tilde{w}_1^k, \dots, \tilde{w}_{N_{\text{calib}}}^k, \tilde{w}_{\text{test}}^k\} \subseteq [0, 1]^{N_{\text{calib}}+1}$ that are tailored to the nominal state/control pair (z_k, v_k) . Starting from (17), we see that $(f(z_k, v_k) - \hat{f}(z_k, v_k)) \in \mathcal{C}(z_k, v_k; s_{1-\alpha}) \oplus \{-\hat{f}(z_k, v_k)\}$, implying that $\varepsilon(z_k, v_k) := f(z_k, v_k) - \hat{f}(z_k, v_k)$ satisfies $\Pr[\varepsilon(z_k, v_k) \in E(z_k, v_k)] \geq 1 - \alpha - \sum_{i=1}^n \tilde{w}_i^k d_{TV}(S^i, S_k)$, where we set $E(z_k, v_k) := \mathcal{C}(z_k, v_k; s_{1-\alpha}) \oplus \{-\hat{f}(z_k, v_k)\}$. In essence, by conformalizing the heuristic uncertainty model Σ , we can obtain a calibrated state-control-dependent model error bound $E(z_k, v_k)$ that holds with high probability.

To construct the normalized weights $\tilde{W}^k$ tailored for a nominal state/control pair (z_k, v_k) , we first define a set of unnormalized weights $W^k = \{w_1^k, \dots, w_{N_{\text{calib}}}^k, w_{\text{test}}^k\} \subseteq \mathbb{R}_{\geq 0}^{N_{\text{calib}}+1}$, where

$$w_i^k = \rho^{\|(z_k, v_k) - (x_i, u_i)\|_2}, \quad (18)$$

where $\rho \in (0, 1]$, for all $(x_i, u_i) \in \mathcal{D}_{\text{calib}}$. The form of (18) encodes the assumption that model error is state-dependent and changes smoothly over the state/control space, with similar state/control pairs having similar errors. Given these unnormalized weights $\tilde{w}_i^k$, we normalize the weights as $\tilde{w}_i^k = \frac{w_i^k}{1 + \sum_{j=1}^n w_j^k}$ to be compatible with the WCP framework (Sec. 3). Then using the Cholesky factor $L(z_k, v_k)$ of $\Sigma(z_k, v_k)$, i.e., $\Sigma(z_k, v_k) = L(z_k, v_k)L(z_k, v_k)^\top$ and defining $s_{1-\alpha}(z_k, v_k) := Q_{1-\alpha_k}(S_k)$, we use (16) and (17) to state that

$$\Pr[\varepsilon(z_k, v_k) \in \underbrace{s_{1-\alpha_k}(z_k, v_k)L(z_k, v_k)\mathcal{B}^{n_x}}_{:=V(z_k, v_k)}] \geq 1 - \alpha - \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i^k d_{TV}(S^i, S_k). \quad (19)$$

While (19) bounds model error at (z_k, v_k) , during execution the system may be perturbed within the reachable tube $\mathcal{R}_k$. Since $(x, u) \neq (z_k, v_k)$ in general, it is important to quantify coverage of $E(z_k, v_k)$ from any $(x, u) \in \mathcal{R}_k$. The ellipsoids $E(z_k, v_k)$ are computed a priori at the nominal point, so we can augment the coverage gap and obtain $\Pr[\varepsilon(x, u) \in E(z_k, v_k)] > 1 - \alpha - 2 \sum_{i=1}^n \tilde{w}_i d_{TV}(S_i, \tilde{S}_k) - \gamma(\mathcal{R}_k)$, where $s_i \sim S_i$, $\tilde{S}_k$ is the distribution of $s_i(z_k, v_k, f(z_k, v_k))$ (the unknown non-conformity score for (z_k, v_k)), and $\gamma(\mathcal{R}_k)$ is defined in Thm. 4. Finally, Thm. 1 derives the probability that $E(z_k, v_k)$ holds for all $(x_k, u_k) \in \mathcal{R}_k$, for all $k \in \{1, \dots, T\}$, solving Prob. 1:

Theorem 1 *Given a length- T trajectory of the true dynamics f (1) $\tau_f \doteq \{(x_k, u_k)\}_{k=1}^T$ such that $(x_k, u_k) \in \mathcal{R}_k$ for all $k \in \{1, \dots, T\}$, desired miscoverage $\{\alpha_k\}_{k=1}^T$, total miscoverage $\sigma_k = \alpha_k + 2 \sum_{i=1}^n \tilde{w}_i d_{TV}(S_i, \tilde{S}_k) + \gamma(\mathcal{R}_k)$, and independence between $\mathcal{D}_{\text{calib}}$ and (x_k, u_k) , we have:*

$$\Pr[\bigcap_{k=1}^T (\varepsilon(x_k, u_k) \in E(z_k, v_k))] \geq 1 - \sum_{k=1}^T \sigma_k := 1 - \tilde{\alpha}. \quad (20)$$

Overall, at every timestep, we predict an ellipsoid $E(z_k, v_k)$ containing the one-step error $\varepsilon(x_k, u_k)$ with high probability. Next, we will use this data-driven, state-dependent bound $E(z_k, v_k)$ in SLS, preserving tractability and shrinking tubes where data is dense while expanding them when OOD.

4.2. Informing SLS-based Planning with CP Model Error Bounds

Using our conformalized model error bounds, we construct an uncertain nonlinear system $z_{k+1} = \hat{f}(z_k, v_k) + V(z_k, v_k)\xi_k$, where $\xi_k \in \mathcal{B}^{n_x}$ and $V(z_k, v_k)\xi_k$ represent the unknown, bounded model error at timestep k . Then we adapt Leeman et al. (2025b) to formulate a nonlinear program (NLP) in (21) to optimize for $\mathbf{z} = \{z_1, \dots, z_T\}$, $\mathbf{v} = \{v_1, \dots, v_T\}$, Φ^x , and Φ^u , given an initial state $\bar{x}_0 \in \mathcal{X}$:

$$\min_{\Phi^x, \Phi^u, \mathbf{z}, \mathbf{v}} J(\mathbf{z}, \mathbf{v}) + \tilde{H}_0(\Phi^x, \Phi^u), \quad (21a)$$

$$\text{s.t. } z_{k+1} = \hat{f}(z_k, v_k), \quad z_1 = \bar{x}_0, \quad \forall k = 1, \dots, T, \quad (21b)$$

$$\Phi_{k+1,j}^x = \mathbf{J}_x \hat{f}(z_k, v_k) \Phi_{k,j}^x + \mathbf{J}_u \hat{f}(z_k, v_k) \Phi_{k,j}^u, \quad (21c)$$

$$\Phi_{j+1,j}^x = V(z_j, v_j), \quad \forall j = 0, \dots, T, \quad \forall k = j+1, \dots, T, \quad (21d)$$

$$\sum_{j=1}^k \|\nabla_x g_i(z_k, v_k) \Phi_{k,j}^x, \nabla_u g_i(z_k, v_k) \Phi_{k,j}^u\|_2 + g_i(z_k, v_k) + b_i + g_i^{\text{lin}} \leq 0, \quad (21e)$$

$$\forall i = 1, \dots, n_c, \quad \forall k = 1, \dots, T.$$

The objective (21a) penalizes the nominal trajectory cost $J(\mathbf{z}, \mathbf{v})$ and the size of reachable tubes $\tilde{H}_0(\Phi^x, \Phi^u)$ (12). Since $\hat{f}$ is more accurate in the training domain, we include an active error reduction cost $J_{\text{active}}(\mathbf{z}, \mathbf{v})$ (defined in App. C.4) that minimizes the distance of $(\mathbf{z}, \mathbf{v})$ to in-distribution calibration points. Constraint (21b) ensures nominal dynamic feasibility, while (21c) and (21d) enforce SLS conditions (10) for the LTV approximation obtained by linearizing $\hat{f}$ around $(\mathbf{z}, \mathbf{v})$, with model error bound $V(z_j, v_j)$ from (19). Constraint (21e) enforces robust state-control satisfaction using disturbance bounds derived from CP. Since SLS guarantees an overapproximation of the reachable tube for a compact disturbance set, (1) the closed-loop trajectory executing $(\mathbf{z}, \mathbf{v}, \Phi^x, \Phi^u)$ over the T -step horizon satisfies (2) with high probability, and (2) safety is maintained with high, though diminishing, probability across replanning steps. The following result solves Prob. 2:

Corollary 2 *Define the forecasted nominal trajectory $\tau_f^{(q)} := \{(z_k^{(q)}, v_k^{(q)})\}_{k=1}^T$ as the T -timestep future nominal trajectory predicted by the learned dynamics $\hat{f}$ at the q th MPC solve (i.e., the $q-1$ th replanning step) in (21b). (a) Then the true closed-loop dynamics (1) under the tracking policy defined by (21) satisfies $\Pr[\bigcap_{k=1}^T ((x_k^{(q)}, u_k^{(q)}) \in \mathcal{F})] \geq 1 - \sum_{k=1}^T \sigma_k^{(q)}$, where $\{(x_k^{(q)}, u_k^{(q)})\}_{k=1}^T$ is the realized*

state-control trajectory if executed for the full T timesteps, and (b) defining $\sigma_1^{(q)}$ as the misscoverage rate for the first forecasted timestep at MPC replanning step q , we have $\Pr[\bigcap_{q=1}^R ((x_1^{(q)}, u_1^{(q)}) \in \mathcal{F})] \geq 1 - \sum_{q=1}^R \sigma_1^{(q)} := 1 - \hat{\alpha}$ for R MPC solves (i.e., $R - 1$ re-planning steps).

Overall, we reduce model error by optimizing an objective that 1) minimizes the volume of closed-loop reachable tubes and 2) penalizes distance to data. Coupling planning with CP-informed reachability ensures high-probability safety for the true dynamics. Finally, refining the CP error set $\mathcal{C}$ with online data (Alg. 1) reduces the coverage gap and improves empirical coverage of model errors.

4.3. Practical Implementation via SCP

To efficiently solve the nonconvex NLP (21), we propose CP-SLS-MPC, described in Alg. 1. CP-SLS-MPC uses sequential convex programming (SCP) (Malyuta et al., 2022). Each SCP iteration is formulated as a second-order cone program (SOCP) (Alg. 1, line 6), obtained by linearizing $\hat{f}$ around the current nominal trajectory $(\mathbf{z}, \mathbf{v})$ and convexifying the constraint tightenings in (21e). Using the solver in Leeman et al. (2024), we iteratively solve a QP for $(\mathbf{z}, \mathbf{v})$ and Riccati recursions for the disturbance feedback matrices (Φ^x, Φ^u) (line 7). For efficiency, we apply a real-time iteration scheme Gros et al. (2020), limiting SCP to one iteration per step (Leeman et al., 2025a), and omit linearization error (Zhan et al., 2025), which is negligible for small time steps (Sec. 6). At each step, we execute the first control input v_1 (line 9), shift the previous solution for initialization (line 5), and re-solve the SCP. Moreover, we improve OOD robustness by updating the error set $\mathcal{V}$ with online observations (line 10), reducing the coverage gap and increasing the likelihood that the closed-loop trajectory remains within the computed reachable tubes.

Algorithm 1 CP-SLS-MPC

```

1: procedure CP-SLS-MPC( $\mathcal{D}_{\text{calib}}$ , horizon  $T$ ,  $\mathcal{F}$ ,  $x_0$ , terminal set  $\mathcal{X}_f$ )
2:    $t \leftarrow 1$  and  $(\mathbf{z}, \mathbf{v}) \leftarrow$  Nominal Solution ▷ Initialize the problem with start state  $\bar{x}_0$ 
3:    $\mathcal{V} \leftarrow \{(x_i, u_i, f(x_i, u_i) - \hat{f}(x_i, u_i))\}_{i=1}^{N_{\text{calib}}}$ , for all  $(x_i, u_i, f(x_i, u_i)) \in \mathcal{D}_{\text{calib}}$  ▷ Initialize error set
4:   while  $x_t \notin \mathcal{X}_f$  do
5:      $\bar{x}_0 \leftarrow x_t$ ; initialize  $(\mathbf{z}, \mathbf{v})$  with shifted previous solution ▷ Initialize with the current state
6:     Linearize (21) around  $(\mathbf{z}, \mathbf{v})$ ; compute  $V(z_k, v_k)$  via (19),  $\mathcal{V}$ ; form SOCP
7:      $(\mathbf{z}_{\text{lin}}^*, \mathbf{v}_{\text{lin}}^*, \Phi^x, \Phi^u) \leftarrow$  SOCP solution ▷ Single iteration of Leeman et al. (2024).
8:     Update  $(\mathbf{z}, \mathbf{v}) \leftarrow (\mathbf{z}, \mathbf{v}) + (\mathbf{z}_{\text{lin}}^*, \mathbf{v}_{\text{lin}}^*)$ .
9:      $u_t \leftarrow v_1$ ;  $x_{t+1} \leftarrow f(x_t, u_t)$  ▷ Closed loop control; state update
10:     $\mathcal{V} \leftarrow \mathcal{V} \cup \{(x_t, u_t, x_{t+1} - \hat{f}(x_t, u_t))\}$ ;  $t \leftarrow t + 1$  ▷ Online data update
11: end procedure

```

5. Theoretical Analysis: Bounding the Coverage Gap

To bound the coverage gap, we assume non-conformity scores from the calibration set $\mathcal{D}_{\text{calib}}$ are independent of the score at the nominal point (z, v) , though they may come from different distributions. We also assume a Lipschitz-type bound on the distribution drift between $\tilde{S}$ and S_i , i.e.,

$$d_{TV}(S_i, \tilde{S}) \leq \epsilon \| [z^\top, v^\top] - [x_i^\top, u_i^\top] \|_2, \quad (22)$$

for $\epsilon \in [0, 1]$, where $s_i \sim S_i$ (15) and $\tilde{S}$ is the non-conformity score distribution of (z, v) . We have:

Theorem 3 *The coverage gap of our weighted conformal predictor follows as,*

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i d_{TV}(S_i, \tilde{S}) \leq 2\epsilon \left[\frac{d_1}{1 - \rho^{d_{\min}}} + \frac{d_{\max} \rho^{d_{\min}}}{(1 - \rho^{d_{\min}})^2} \right], \quad (23)$$

where we sort the distance to points in the calibration set such that $d_i = \| [z^\top, v^\top] - [x_i^\top, u_i^\top] \|_2$, $d_1 \leq d_2 \leq \dots \leq d_n$, $d_{\min} = \min_{1 \leq i \leq n-1} \{d_{i+1} - d_i\}$, and $d_{\max} = \max_{1 \leq i \leq n-1} \{d_{i+1} - d_i\}$.

Thm. 3 shows that the gap depends on state-control calibration point density. Greater point density decreases $d_{\min}$, increasing the gap; reducing ρ mitigates this, but doing so can cause the $(1 - \alpha)$ -quantile to diverge. The term $\frac{d_1}{1 - \rho d_{\min}}$ shows that being closer to the training data reduces the gap, as d_1 is the minimal distance between $\mathcal{D}_{\text{calib}}$ and (z, v) . Online augmentation of $\mathcal{V}$ may slightly violate independence (due to time-correlation), but for large N_{calib} , Thm. 3 suffices. As SLS requires model error to hold for all $(x, u) \in \mathcal{R}$, we also provide a coverage bound valid over the reachable tube: **Theorem 4** *Under the same distributional assumption of Theorem 3, we have that for all $(x, u) \in \mathcal{R}$ – the tube around the nominal point (z, v) , $\Pr[\varepsilon(x, u) \in E(z, v)] \geq 1 - \alpha - 2 \sum_{i=1}^n \tilde{w}_i d_{TV}(S_i, \tilde{S}) - \gamma(\mathcal{R})$, for $\gamma(\mathcal{R}) := 2\hat{\epsilon}M(\mathcal{R})$, where $M(\mathcal{R})$ is the maximum axis length of tube $\mathcal{R}$, and $\hat{\epsilon} \leq \epsilon$ is chosen such that (22) holds for all $(x, u) \in \mathcal{R}$.*

Using Theorem 4, we find that the coverage guarantees remain close to the desired coverage level α , assuming a gradual spatial-drift of the error distribution. Furthermore, the optimization problem (21) seeks to minimize the tube volume in the cost function, thereby minimizing $\gamma(\mathcal{R})$.

6. Results

We evaluate our approach on robust planning with 4D Dubins’ and 12D quadcopter models. Comparisons include vanilla MPC (V-MPC) on the learned dynamics $\hat{f}$ without robust constraint tightenings and a fixed hyper-ball variant of our method (CP-Ball), where $\Sigma(z_k, v_k)$ in (15) is replaced with I_{n_x} , highlighting the benefit of state-dependent CP-Ellipsoid uncertainty sets. Dynamics models include a 4D Dubins’ car (2 controls) and a 12D quadcopter (4 controls) (see App. B.1, B.2). Training details, model architectures, and cost functions are in App. C. Implementation uses PyTorch, Acados, and L4CasAdi. Experiments run on an M4 Pro MacBook (12 cores, 24 GB RAM), reporting average MPC step time, prediction error $|f(x, u) - \hat{f}(x, u)|$, and minimum obstacle distance. All trials use horizon $T = 15$ with $\alpha_k = 0.1/15$ per timestep.

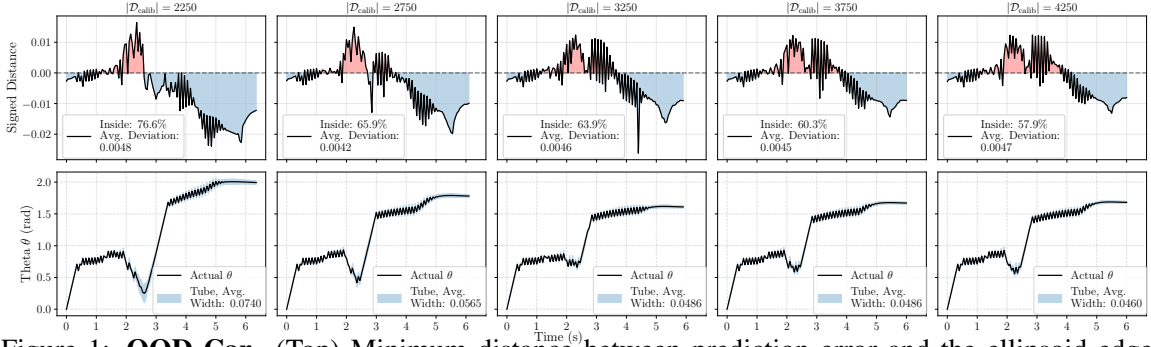


Figure 1: **OOD Car.** (Top) Minimum distance between prediction error and the ellipsoid edge is plotted for varying D_{calib} with a fixed start and goal; negative values indicate points inside the ellipsoid. (Bottom) Corresponding one-step θ tubes are shown. Smaller D_{calib} keeps prediction errors inside the ellipsoid more often and results in larger tubes.

Dubins’: In Domain Performance. To validate our approach, we used the 4D Dubins’ car model described in (39) in App. B.1. In this experiment, we select start and goal states such that the car’s states would remain within the training distribution throughout the rollout. We define $\mathcal{F} = \{(x, u) \mid p_x \in [0, 5], p_y \in [-5, 5], \|[p_x, p_y]^\top - [2.5, 0]\|_2 \geq 1\}$, with bounds on p_x and p_y and a 1 meter obstacle centered at $(2.5, 0)$. Across 10 experiments with varying start and goal states, we found that all three approaches: V-MPC, CP-Ball, and CP-Ellipsoid were successful in reaching the goal while avoiding the obstacle. In Table 1, we found that V-MPC was faster, since the other approaches spend time on forward error propagation and constraint tightening. The obstacle proximity increases with CP-Ball and CP-Ellipsoid—a byproduct of the constraint tightening (Table 1). Finally, the one-step

prediction error for $\hat{f}$ remains inside the ball and ellipsoid over $93.33\% = (1 - \frac{0.1}{15}) \times 100\%$, thus validating the coverage guarantees (Fig. 5 in App. D). Within the training domain, we expect V-MPC to have similar success rates as our method since $\hat{f}$ is accurate; this changes in OOD settings.

Dubins’: OOD Performance. Next, to evaluate performance fully out of distribution (OOD), we train a dynamics and uncertainty model such that $p_y \in [-12, -6] \cup [6, 12]$ and modify the dynamics to gradually steer the car’s angle toward the obstacle for $p_y \in [-6, 6]$ (discussed in App. B.1), while retaining the obstacle configuration from the in-domain experiment. In this setting, we vary the calibration set size from 2,250 to 4,250 points for CP-Ellipsoid and ensure that the start and goal positions lie outside the training domain. As shown in Fig. 1, the one-step prediction errors initially lie outside the predicted ellipsoid but gradually return within it as online calibration points are collected. Moreover, with fewer initial calibration points, the online adaptation occurs more rapidly, resulting in a higher coverage of errors in the ellipsoid. The plot also shows that the one-step tubes in θ remains wider for smaller $\mathcal{D}_{\text{calib}}$. Since the car successfully reaches the goal, these results suggest that the SLS tube over-approximation, when coupled with conformal prediction, enables robust OOD performance.

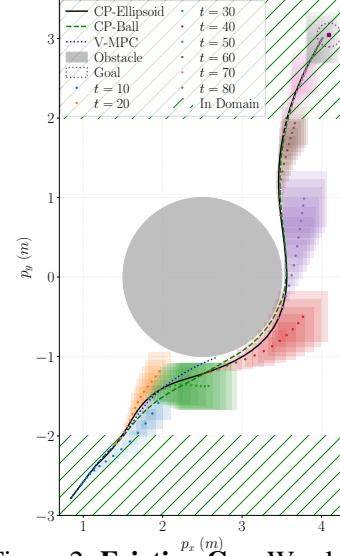


Figure 2: **Friction Car.** We plot all approaches and forecasted MPC steps & tubes for CP-Ellipsoid.

Dubins’: Disjoint Training Domains (Friction Car).

To evaluate OOD obstacle avoidance under limited online data, we contruct disjoint training regions (similar to the OOD setup) where $p_y \in [-5, -2] \cup [2, 5]$. The start and goal positions are randomly sampled from different regions, ensuring that all feasible trajectories must traverse OOD areas to reach the goal. In the OOD region, we apply the same steering modification as in the previous experiment, while adding p_y dependent acceleration dynamics (see App. B.1 for details). Across 10 runs, V-MPC approach failed twice, the CP-Ball failed once, and CP-Ellipsoid was consistently successful. These differences arise from reduced obstacle clearance (Table 1) observed in the CP-Ball and V-MPC cases. Figure 3 shows that both tube-based methods reach the goal, but CP-Ellipsoid maintains a greater distance from the obstacle, as evident from its forecasted trajectories and uncertainty tubes. Fig. 3 also illustrates that the state-dependent ellipsoids expand along the direction of motion, as increased forecasted y -motion leads to higher y -direction uncertainty. A similar result from an extra run is in App. D.

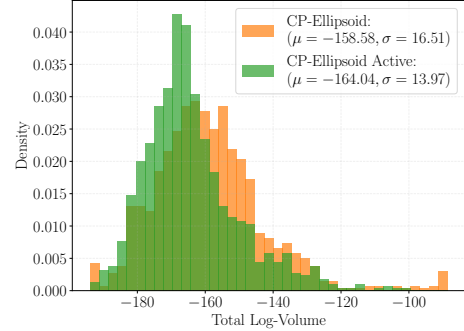
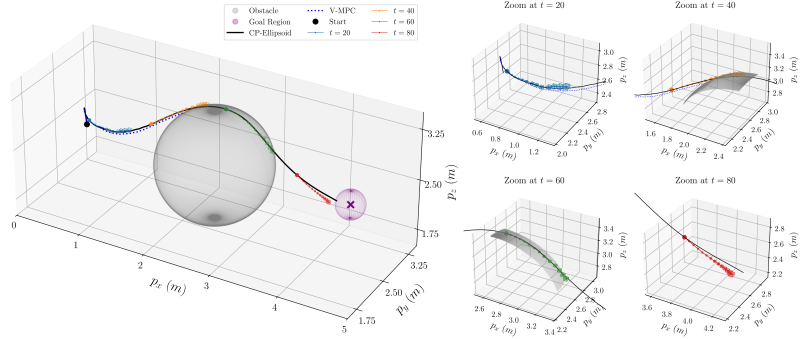


Figure 3: **Active Uncertainty.** We plot the histogram of summed log-volumes for each forecasted MPC plan. With active uncertainty reduction, the distribution shifts left, indicating lower encountered uncertainty during rollouts.

Dubins’: Active Uncertainty Reduction. Finally, to evaluate active uncertainty reduction, we trained $\hat{f}$ and Σ without sampling data in a circular region centered at $(p_x, p_y) = (2.5, 0.0)$ with a radius of 1 meter, and introduced friction and attractive steering dynamics within this unseen region.

Figure 4: **Quadcopter.** A rollout of the quadcopter with V-MPC and CP-Ellipsoid. CP-Ellipsoid maintains sufficient proximity to avoid crashing into the obstacle while V-MPC crashes.



Since no obstacle lies between the start and goal states, the trajectory optimizer may plan through this area. Despite this, the active uncertainty reduction method achieves lower prediction error by planning a longer path that remains in the training domain (Table 2 App. D). Moreover, Fig. 3 shows the log-volume of the uncertainty tubes, showing that the active uncertainty cost ((50) in App. C.4) yields smaller tubes (Fig. 3, green) compared to the baseline without this cost (Fig. 3, orange).

Table 1: Mean, stdev of computation times and prediction errors and the mean minimum obstacle distance for vanilla-MPC (V-MPC), the CP-Ball (Ball), and CP-Ellipsoid (Ellipsoid).

Model	Computation Time (ms)			Prediction Error (L_2) ($\times 10^{-2}$)			Min. Obstacle Dist. (m)		
	V-MPC	Ball	Ellipsoid	V-MPC	Ball	Ellipsoid	V-MPC	Ball	Ellipsoid
Car	53.3 ± 2.1	126.6 ± 8.4	126.7 ± 8.3	0.28 ± 0.54	0.26 ± 0.14	0.26 ± 0.14	1.152	1.202	1.207
Friction Car	53.7 ± 2.6	131.1 ± 1.4	129.7 ± 9.0	1.94 ± 2.87	1.92 ± 2.80	1.95 ± 2.69	1.315	1.376	1.464
Quadcopter	63.1 ± 4.6	nan $\pm$ nan	222 ± 20.9	3.89 ± 5.49	nan $\pm$ nan	3.30 ± 4.17	0.804	nan	0.809

12D Quadcopter To evaluate the scalability of our framework to higher-dimensional systems, we train a 12D quadcopter dynamics model. Using this model, we plan trajectories such that the spatial coordinates (p_x, p_y, p_z) remain outside a sphere centered at $(2.5, 2.5, 2.5)$ with a radius of 0.8 meters – leaving a 0.2-meter-thick shell around the sphere that is out of distribution and characterized by increased downward acceleration. To improve the performance of the uncertainty model, we modify its weighting by blending the covariance matrix with the identity matrix to reduce conservativeness, i.e., using $((1 - \tau)I + \tau\Sigma(z, v))$ instead of $\Sigma(z, v)$ in (16). With this approach, the CP-Ellipsoid method successfully planned 8 out of 10 trajectories (one failure was due to numerical instability, which did not result in collision), whereas V-MPC approach succeeded in only 4 out of 10 runs. While this demonstrates improved robustness over V-MPC, the failed case suggests faster tube expansion may be necessary to ensure consistent collision avoidance. Finally, Fig. 4 illustrates a successful quadcopter trajectory, demonstrating that our CP-informed tubes maintain safe proximity to the obstacle.

7. Conclusion

In this work, we developed an efficient framework for robust out-of-distribution MPC with learned dynamics, using WCP and SLS. We provided theoretical guarantees ensuring that the rollout satisfy constraints with high probability, and introduced an uncertainty reduction cost function to encourage the system to remain within the training domain when possible. Through experiments on variants of the 4D Dubins car and 12D quadcopter, we demonstrated that the CP-SLS-MPC framework achieves robust performance even when using learned dynamics that generalize poorly OOD.

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Appendix A. Proofs:

A.1. Proof of Theorem 3 and Corollary 5

Proof (Theorem 3) From Theorem 2 and Lemma 1 in Barber et al. (2023), the coverage gap for independent data is bounded by:

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot d_{TV}(S, S^i) \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot 2d_{TV}(S_i, \tilde{S}) \quad (24)$$

Given the normalized weights are $\tilde{w}_i = \frac{w_i}{1 + \sum_{j=1}^{N_{\text{calib}}} w_j}$, and $w_i = \rho^{d_i}$ for $d_i = \|[z^\top, v^\top] - [x_i^\top, u_i^\top]\|_2$

We can substitute into Equation 24:

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \left(\frac{\rho^{d_i}}{1 + \sum_{j=1}^{N_{\text{calib}}} \rho^{d_j}} \right) \cdot 2\epsilon \cdot d_i \quad (25)$$

Without loss of generality, we can rearrange the sum to create a decaying series $\{\rho^{d_i}\}_{i=1}^{N_{\text{calib}}}$, where d_i is increasing. While this series is not purely geometric, we can bound it by two geometric series. Let $d_{\min} = \min\{1 \leq i \leq N_{\text{calib}} - 1 | d_{i+1} - d_i\}$ and $d_{\max} = \max\{1 \leq i \leq N_{\text{calib}} - 1 | d_{i+1} - d_i\}$. Then, $\rho^{d_1} \rho^{d_{\max}(i-1)} \leq \rho^{d_i} \leq \rho^{d_1} \rho^{d_{\min}(i-1)}$. Thus, we can bound Equation 25,

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \left(\frac{\rho^{d_i}}{1 + \sum_{j=1}^{N_{\text{calib}}} \rho^{d_j}} \right) \cdot 2\epsilon \cdot d_i \leq 2\epsilon \left(\frac{\sum_{i=1}^{N_{\text{calib}}} \rho^{d_1} \rho^{d_{\min}(i-1)} \cdot (d_1 + d_{\max}(i-1))}{1 + \sum_{j=1}^{N_{\text{calib}}} \rho^{d_1} \rho^{d_{\max}(j-1)}} \right)$$

Then observe the sum of the numerator,

$$N_{\text{sum}} = \sum_{i=1}^{N_{\text{calib}}} \rho^{d_1} \rho^{d_{\min}(i-1)} \cdot (d_1 + d_{\max}(i-1)) \quad (26)$$

$$= \rho^{d_1} \left[d_1 \left(\frac{1 - \rho^{N_{\text{calib}} \cdot d_{\min}}}{1 - \rho^{d_{\min}}} \right) + d_{\max} \left(\frac{\rho^{d_{\min}} - N_{\text{calib}} \rho^{n \cdot d_{\min}} + (N_{\text{calib}} - 1) \rho^{(N_{\text{calib}} + 1) d_{\min}}}{(1 - \rho^{d_{\min}})^2} \right) \right] \quad (27)$$

Similarly, the sum the of the denominator,

$$D_{\text{sum}} = 1 + \sum_{j=1}^{N_{\text{calib}}} \rho^{d_1} \rho^{d_{\max}(j-1)} \quad (28)$$

$$= 1 + \rho^{d_1} \left(\frac{1 - \rho^{N_{\text{calib}} \cdot d_{\max}}}{1 - \rho^{d_{\max}}} \right) \quad (29)$$

Then, the coverage gap is bounded by,

$$\text{Coverage Gap} \leq 2\epsilon \frac{N_{\text{sum}}}{D_{\text{sum}}} \quad (30)$$

$$\leq 2\epsilon \left[d_1 \left(\frac{1 - \rho^{N_{\text{calib}} \cdot d_{\min}}}{1 - \rho^{d_{\min}}} \right) + d_{\max} \left(\frac{\overbrace{\rho^{d_{\min}} - N_{\text{calib}} \rho^{N_{\text{calib}} \cdot d_{\min}} + (N_{\text{calib}} - 1) \rho^{(N_{\text{calib}} + 1) d_{\min}}}^{\leq 0}}{(1 - \rho^{d_{\min}})^2} \right) \right] \underbrace{\frac{1 - \rho^{d_{\max}}}{1 - \rho^{N_{\text{calib}} \cdot d_{\max}}}}_{\leq 1} \quad (31)$$

$$\leq 2\epsilon \left[\frac{d_1}{1 - \rho^{d_{\min}}} + \frac{d_{\max} \rho^{d_{\min}}}{(1 - \rho^{d_{\min}})^2} \right] \quad (32)$$

■

For large N_{calib} , (i.e., as $N_{\text{calib}} \rightarrow \infty$), we can achieve a tighter upper bound

Corollary 5 *Under the same assumptions of Theorem 3 we have that as $N_{\text{calib}} \rightarrow \infty$ the coverage gap is more tightly bounded by,*

$$\text{Coverage Gap} \leq 2\epsilon \left[\frac{d_1}{1 - \rho^{d_{\min}}} + \frac{d_{\max} \rho^{d_{\min}}}{(1 - \rho^{d_{\min}})^2} \right] (1 - \rho^{d_{\max}}) \quad (33)$$

Proof (Corollary 5) Taking Equation 31 as $n \rightarrow \infty$ we complete the proof. ■

A.2. Proof of Theorem 4

Proof (Theorem 4) Using Equation 24, we have that

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot d_{TV}(S, S^i) \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot 2d_{TV}(S_i, \hat{S}), \quad (34)$$

where $\hat{S}$ is the distribution of the observed state and control (x, u) . Then, using the triangle inequality we have that,

$$\text{Coverage Gap} \leq \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot d_{TV}(S, S^i) \leq \underbrace{\sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot 2d_{TV}(S_i, \tilde{S})}_{(*), \text{Bounded in Eq. 32}} + \underbrace{\sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot 2d_{TV}(\tilde{S}, \hat{S})}_{(**)}. \quad (35)$$

Observing that the left sum (*) is already bounded by Theorem 3, we proceed to bound the right sum (**). Then, realizing the sum of normalized weights is ≤ 1 , we obtain,

$$\sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \cdot 2d_{TV}(\tilde{S}, \hat{S}) = 2d_{TV}(\tilde{S}, \hat{S}) \sum_{i=1}^{N_{\text{calib}}} \tilde{w}_i \leq 2d_{TV}(\tilde{S}, \hat{S}) \quad (36)$$

Then, using the Lipschitz-type bound,

$$2d_{TV}(\tilde{S}, \hat{S}) \leq 2\hat{\epsilon} \| [x^\top, u^\top] - [z^\top, v^\top] \|_2 \leq 2\hat{\epsilon} M(\mathcal{R}) =: \gamma(\mathcal{R}), \quad (37)$$

where, $(x, u) \in \mathcal{R}$, $M(\mathcal{R})$ is the maximum axis length of the tube, and $\hat{\epsilon} \leq \epsilon$ is the local Lipschitz constant in the tube. The remaining proof follows substituting the bound, $\gamma(\mathcal{R})$ for (**) in (35). ■

A.3. Proof Sketch of Theorem 1 and Corollary 2

Proof Sketch (Theorem 1) Using Theorem 4 we get the probability of error coverage for each step in the rollout. Using the approach of Proposition 2 and Theorem 1 from Chee et al. (2024) and Lindemann et al. (2023) respectively we complete the proof. ■

Proof Sketch (Corollary 2) (a) Since SLS provides guarantees conditioned on valid disturbance bounds, Theorem 1 directly implies the result in Corollary 2(a).

(b) Instead of applying the union bound approach across the forecasted MPC plan, we bound the intersection probability of the first-step errors remaining in the predicted ellipsoids for all R planning steps using the same approach as Theorem 4 we get:

$$\Pr\left[\bigcap_{k=1}^T \varepsilon(z_1^{(q)}, v_1^{(q)}) \in E(z_1^{(q)}, v_1^{(q)})\right] \geq 1 - \sum_{q=1}^R \sigma_1^q, \quad (38)$$

Notably, we omit the tubes since the tubes at the first step of each replan, $\mathcal{R}_1^q = \{(z_1^{(q)}, v_1^{(q)})\}$ since the planner knows the initial state. Then, using the same approach as (a) we get the result in (b). ■

Appendix B. Dynamics Models

Below we discuss the dynamics models used in our experiments. We provide the continuous time dynamics which were used to train our models. We obtained the discrete model by using Euler Discretization.

B.1. Car Dynamics Models

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \cos(\theta) \\ v \sin(\theta) \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \omega \\ a \end{bmatrix} \quad (39)$$

Equation 39 describes the control-affine dynamics models for Dubins' car, where p_x and p_y are the car's x and y positions. θ and v are the direction and velocity of the car. Lastly, the control inputs directly steer (ω) and accelerate (a) the car. The dynamics model in Equation 39 was used strictly for the in-distribution experiments.

OOD Experiment Dynamics For the out of distribution (OOD) Dubins' car experiment we modified the θ dynamics by adding attractive steering towards the obstacle centered at (2.5, 0) in the region without training data,

$$\dot{\theta} = \omega_{\text{attr}} \mathbf{1}[|p_y| < 6] + \omega \quad (40)$$

$$d^2 = (p_x - 2.5)^2 + p_y^2 \quad (41a)$$

$$\theta_a = \text{atan2}(p_y, p_x) \quad (41b)$$

$$\Delta\theta_{\text{norm}} = \text{atan2}(\sin(\theta_a - \theta), \cos(\theta_a - \theta)) \quad (41c)$$

$$\omega_{\text{attr}} = \left(\frac{k_{\text{attr}}}{d^2}\right) \Delta\theta_{\text{norm}}, \quad (41d)$$

where the attractive steering dynamics are defined in Equation 41, and atan2 is the arctangent function accounting for the x-y quadrant, and $\mathbf{1}[\cdot]$ is the indicator function. For the OOD experiment we set k_{attr} , the strength of the attractive force to -0.5 (positive values repel). Notably, the strength of the increases as car gets closer to the obstacle.

Active Uncertainty Experiment Dynamics Similar to the OOD dynamics we include attractive steering (with $k_{\text{attr}} = -0.5$, we modify the the v dynamics of the car such that,

$$\dot{\theta} = \tilde{\omega}_{\text{attr}} \mathbf{1}[d^2 < 1] + \omega \quad (42a)$$

$$\dot{v} = -(1 + \cos(\pi\sqrt{d^2})) \mathbf{1}[d^2 < 1] + a \quad (42b)$$

$$\tilde{\omega}_{\text{attr}} = \left(\frac{k_{\text{attr}}}{d^2 + 0.1} \right) \Delta\theta_{\text{norm}}, \quad (42c)$$

where d^2 and $\Delta\theta_{\text{norm}}$ are defined in Equation 41. We modify the denominator of ω_{attr} to prevent a singularity at (2.5, 0.0)—this was not a problem in the OOD case since there was an obstacle present at (2.5, 0.0). The augmented velocity dynamics add a friction term to reduce the speed of the car in this region.

Friction Car Dynamics Lastly, we modified Dubins' car such that the velocity dynamics were modified over the whole state-space, while including attractive steering (with $k_{\text{attr}} = -1.5$):

$$\dot{\theta} = \omega_{\text{attr}} \mathbf{1}[|p_y| < 2] + \omega \quad (43a)$$

$$\dot{v} = a - \text{sign}(v) \left(0.1 \cos\left(\frac{2\pi}{5} p_y\right) + 0.1 - a_{\text{slip}} \mathbf{1}[|p_y| < 2] \right) \quad (43b)$$

$$a_{\text{slip}} = 0.7e^{-2p_y^2} \quad (43c)$$

In this model, we include a slippage term to increase the velocity of the car for $-2 < p_y < 2$, while providing a changing friction term over the whole state-space.

B.2. 12D Quadcopter Dynamics Model

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \\ \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \\ \ddot{p}_x \\ \ddot{p}_y \\ \ddot{p}_z \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = f(x, u) := \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \\ q \sin(\phi) / \cos(\theta) + r \cos(\phi) / \cos(\theta) \\ q \cos(\phi) - r \sin(\phi) \\ p + q \sin(\phi) \tan(\theta) + r \cos(\phi) \tan(\theta) \\ \frac{u_1}{m} (\sin(\phi) \sin(\psi) + \cos(\phi) \cos(\psi) \sin(\theta)) \\ \frac{u_1}{m} (\cos(\psi) \sin(\phi) - \cos(\phi) \sin(\psi) \sin(\theta)) \\ g + \frac{u_1}{m} (\cos(\phi) \cos(\theta)) - a_{\text{fall}} \\ \frac{I_y - I_z}{I_x} qr + \frac{u_2}{I_x} \\ \frac{I_z - I_x}{I_y} pr + \frac{u_3}{I_y} \\ \frac{I_x - I_y}{I_z} pq + \frac{u_4}{I_z} \end{bmatrix} \quad (44)$$

$$a_{\text{fall}} = (0.1 + 0.1 \cos(2\pi d^2 - \pi)) \mathbf{1}[d^2 < 1] \quad (45)$$

$$d^2 = (p_x - 2.5)^2 + (p_y - 2.5)^2 + (p_z - 2.5)^2 \quad (46)$$

For our experiments, we used the 12D Quadcopter dynamics model described in Equation B.2 (Sabatino, 2015). p_x, p_y , and p_z describe the x, y, and z positions of the quadcopter while $\dot{p}_x, \dot{p}_y$, and $\dot{p}_z$ describe the x, y, and z velocities. ψ, θ , and ϕ are the roll, pitch, and yaw. Similarly, p, q , and r are the roll rate, pitch rate, and yaw rates, respectively are directly controlled by inputs u_2, u_3 , and u_4 . The control input, u_1 , directly impacts the thrust-z-direction acceleration. Lastly, the constants g, m, I_x, I_y , and I_z are the acceleration due to gravity, the mass of the quadcopter, and the principle moments of inertia. For our experiments, we let $g = -9.81m/s^2$, $m = 1.0kg$, $I_x = 0.5kg \cdot m^2$, $I_y = 0.1kg \cdot m^2$, and $I_z = 0.3kg \cdot m^2$. Notably, the expression for $\ddot{p}_z$ includes a_{fall} to introduce a falling term for OOD dynamics. The falling dynamics are not encounter while training the model as described in App. C.1.

Appendix C. Experiment Setup

C.1. Data Sampling

Dubins' Car & Friction Car: To train the in-distribution Dubins' car dynamics and uncertainty models, we applied uniform sampling to the following state-control space:

$$[p_x, p_y, \theta, v] \times [\omega, a] \in [0, 5] \times [-5, 5] \times [-\pi, \pi] \times [-10, 10] \times [-10, 10] \times [-10, 10]. \quad (47)$$

Similarly, for the out-of-distribution (OOD) car, we instead sampled $p_y \in [-12, -6] \cup [6, 12]$ with the remaining state and controls remaining the same as Equation 47. For the friction car model, we sampled $p_y \in [-5, -2] \cup [2, 5]$, while using Equation 47 for the remaining dimensions. Lastly, for our active uncertainty experiments, we also sampled from Equation 47, and enforced that $(p_x - 2.5)^2 + (p_y)^2 > 1$ to create an OOD region. For both the OOD car and active uncertainty experiments, we ensured our learned model was restricted to training on areas where modifications did not impact the car dynamics model.

Using the sampled points, we computed the ground truth using the expert dynamics model, as formulated in Equation 39, to create our dynamics training dataset (1,000,000 points), uncertainty training dataset (1,000,000 points), and calibration datasets (varying numbers of points).

12D Quadcopter: To get informative data points to train our neural-network model, we collected expert trajectories such that the states satisfied,

$$p_x, p_y, p_z \in [0, 5] \text{ s.t. } (p_x - 2.5)^2 + (p_y - 2.5)^2 + (p_z - 2.5)^2 > 1 \quad (48a)$$

$$\psi, \theta, \phi \in [-\pi/4, \pi/4] \quad (48b)$$

$$\dot{p}_x, \dot{p}_y, \dot{p}_z \in [-5, 5] \quad (48c)$$

$$p, q, r \in [-2, 2] \quad (48d)$$

$$u_1 \in [0, -2gm], g = -9.81m/s^2, m = 1.0kg \quad (48e)$$

$$u_2, u_3, u_4 \in [-0.1, 0.1] \quad (48f)$$

to create an OOD region centered at (2.5, 2.5, 2.5) with a radius of 1. Additionally, the goal position was $[p_x = 4.5, p_y = 2.5, p_z = 2.5]$, while the start positions was arbitrarily sampled from Equation 48. Our discrete timestep was $0.05s$ and our trajectory length was 100 timesteps. We sampled 100,000 trajectories and included state-control pairs from trajectories that the optimizer (IpOpt) was able to solve. This resulted in a dynamic training and uncertainty-trained set with 4,702,400 points each, leaving up to 100,000 points for the calibration dataset.

C.2. Model Architecture and Training

For our learned model (dynamics and uncertainty), we use an MLP with one hidden layer and the tanh activation function. Our uncertainty model was trained to output Cholesky factors, ensuring the matrices were positive-definite and facilitating faster performance in our optimization problem.

For Dubins' car and the friction car dynamics models, we used 4,096 and 2,048 hidden nodes for the dynamics and uncertainty models, respectively. Similarly, for the 12D quadcopter, we used 1,024 hidden nodes for both the dynamics and uncertainty models, and were trained for 200 epochs. Lastly, we used a learning rate of 10^{-4} and 10^{-5} for the dynamics and uncertainty models, respectively.

C.3. Optimization Parameters and Cost Functions

For the SLS forward propagation and constraint tightenings, we used $\bar{P}^{\frac{1}{2}} = 10^6 I_{n_x}$, $\bar{Q}^{\frac{1}{2}} = 10^6 I_{n_x}$ and $\bar{R}^{\frac{1}{2}} = 10^6 I_{n_u}$, as found in Equation 12. For trajectory optimization, we considered the following LQR cost,

$$J(\mathbf{z}, \mathbf{v}) = (z_T - z_{\text{goal}})^\top Q_f (z_T - z_{\text{goal}}) + \sum_{k=1}^{T-1} \left(z_k^\top Q z_k + (z_{k+1} - z_k)^\top Q_s (z_{k+1} - z_k) + u_k^\top R u_k \right), \quad (49)$$

where $Q_f, Q_s, Q \in \mathbb{R}^{n_x \times n_x}$ and $R \in \mathbb{R}^{n_u \times n_u}$ are symmetric positive-definite matrices, and z_{goal} is the goal state. For all dynamic models, we used diagonal matrices, represented as $\text{diag}(a_1, a_2, \dots, a_n)$ and $\text{block_diag}(\dots)$ representing block diagonal matrices. For Dubins' car (excluding the active uncertainty experiment) and the friction car, we let $Q_f = \text{diag}(1, 1, 0, 1)$, $Q = Q_s = 0_{4 \times 4}$ (the 4×4 zero matrix), and $R = \text{diag}(0.1, 0.1)$. For the 12D Quadcopter, we set $Q_f = \text{block_diag}(I_3, 0_{3 \times 3}, I_3, 0.2I_3)$, $Q = \text{block_diag}(0_{3 \times 3}, 0_{3 \times 3}, 0.01I_3, 0_{3 \times 3})$, $Q_s = 0_{12 \times 12}$, and $R = \text{diag}(0.01, 0.1, 0.1, 0.1)$.

Lastly, regarding conformal prediction parameters, we used $\rho = 0.97$ for the car model and $\rho = 0.8$ for the quadcopter model, respectively. Outside of the OOD car experiment, we used 10,000 calibration points for the car models and 30,000 calibration points for the quadcopter model. For all the models we set $\alpha_k = \frac{0.1}{15}$ s.t. the sum total of α would be 0.1.

C.4. Active Uncertainty

To enable active uncertainty reduction, we sought to minimize the distance to the positional-state variables (i.e., p_x and p_y) from $\mathcal{D}_{\text{calib}}$. To achieve efficient performance, we used K-means clustering [Hastie et al. \(2009\)](#) to find a representative set of L points, $\tilde{\mathcal{D}} = \{(p_{x_i}, p_{y_i})\}_{i=1}^L$. Then, we minimized the following cost for Dubins' car,

$$J_{\text{active}}(\mathbf{z}, \mathbf{v}) = \sum_{k=0}^{T-1} \exp \left[-\frac{1}{L+G} \left(G e^{-\beta \| (p_{x_{\text{goal}}, p_{y_{\text{goal}}, v_{\text{goal}}}) - (p_{x_k, p_{y_k}, v_k}) \|^2} + \sum_{j=1}^L e^{-\beta \| (p_{x_j, p_{y_j}}) - (p_{x_k, p_{y_k}}) \|^2} \right) \right], \quad (50)$$

where $p_{x_{\text{goal}}}, p_{y_{\text{goal}}}$, and v_{goal} are the goal x-position, y-position, and velocities, p_{x_k}, p_{y_k} , and v_k are the x-position, y-position, and velocities of the nominal point z_k while p_{x_i} , and p_{y_i} are the x and y positions of points from $\tilde{\mathcal{D}}$, and G is a weight for the goal node (effectively creating G points at

the goal). Lastly, β is a sharpness parameter to ensure that we minimize the distance to a particular point in $\tilde{\mathcal{D}}$ as opposed to minimizing the distance to all points. The inclusion of the goal point enabled us to provide a high weight to the cost without hinder the MPC optimizer from reaching the goal. Since our goal is to minimize the distance to the representative points by minimizing the cost function we scale the internal sum by -1 . For efficient implementation we included this as a part of the quadratic cost, thus requiring the outer-exponential to provide a lower-bound to the cost. In our experiments we choose $L = 800$, $G = 200$, and $\beta = 3$, and our total cost was $J_{\text{total}}(\mathbf{z}, \mathbf{v}) = J(\mathbf{z}, \mathbf{v}) + 1000J_{\text{active}}(\mathbf{z}, \mathbf{v})$. For Q_f , Q , and R we used the same parameters as the remaining Dubins' car experiments and let $Q_s = 0.1I_4$.

Appendix D. Additional Results

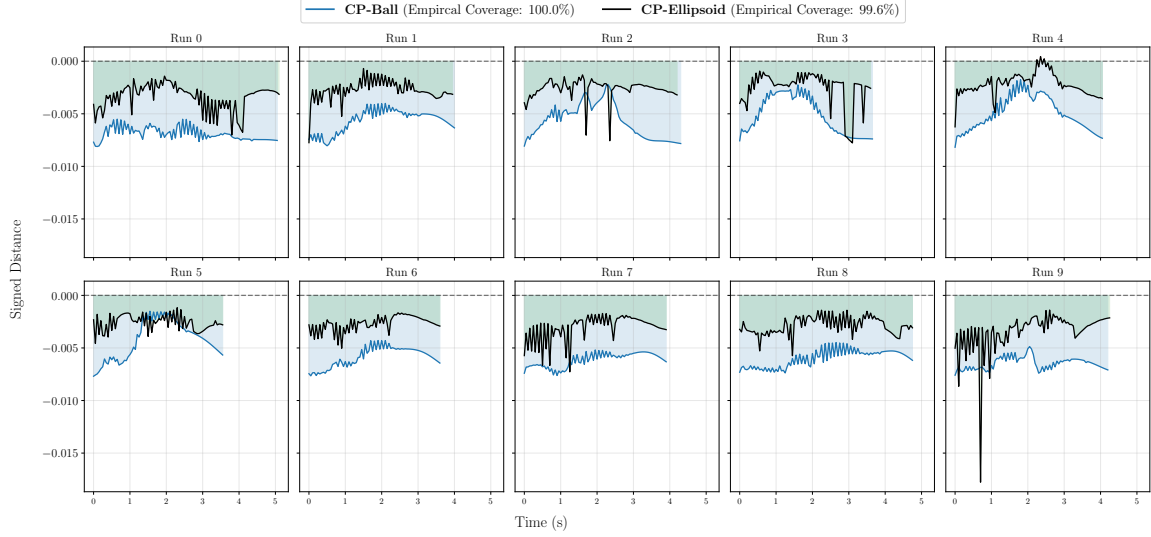


Figure 5: **In Domain Car.** We plot the minimum distance of the prediction error to the edge of the ellipsoid. Both the fixed ball and adaptive ellipsoid method empirically satisfy the coverage guarantee and remain inside over $99.33\% = (1 - \frac{0.1}{15}) \times 100\%$ of the time.

Table 2: **Active Uncertainty.** The computation times (ms), and prediction errors when planning with vanilla-MPC, the adaptive ellipsoid with and without active uncertainty.

Model	V-MPC	CP-Ellipsoid	CP-Ellipsoid + Active Uncertainty
Time (ms)	56.1 ± 1.1	132.0 ± 9.9	131.4 ± 8.2
Pred. Err. (L_2) ($\times 10^{-2}$)	2.82 ± 7.85	2.83 ± 7.89	$.34 \pm .59$

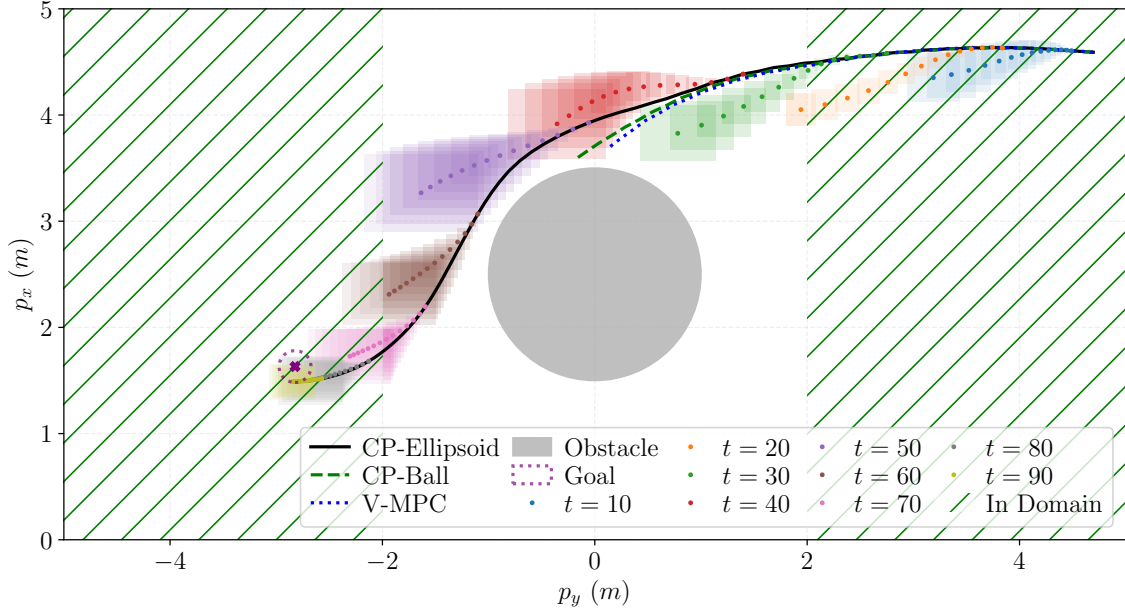


Figure 6: **Friction Car.** We provide an example run of the friction area where the adaptive ellipsoid approach is successful in avoiding the obstacle. The remaining approaches fail because they do not maintain sufficient obstacle proximity and thus crash into the obstacle.